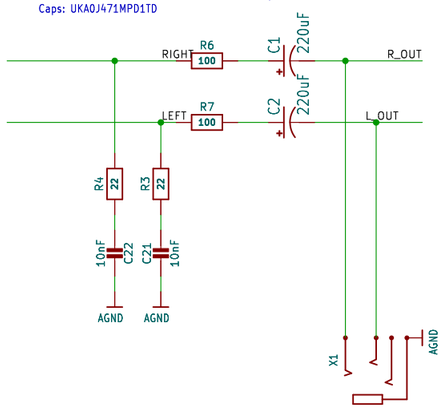
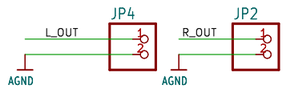
C1, C2 = 1 / (2*π*f*Xc)

16 ohm load at 20Hz cuttoff 1 / (6.2832*20hz*16) = 497µF

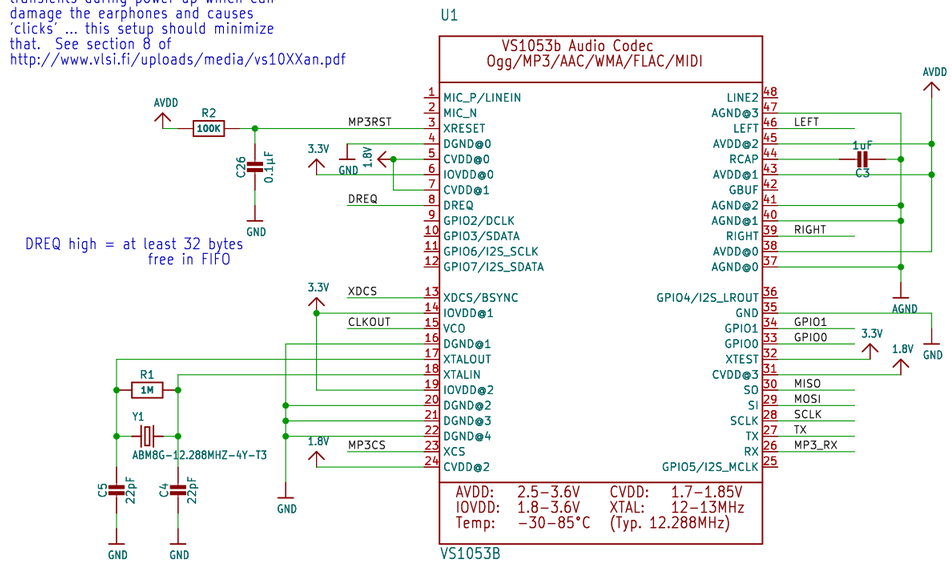
32 ohm load at 20Hz cuttoff 1 / (6.2832*20hz*32) = 249µF

Caps: UKA0J471MPD1TD





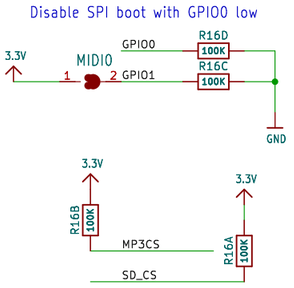
Enabling the IC too soon can cause transients during power up which can damage the earphones and causes 'clicks' ... this setup should minimize that. See section 8 of http://www.vlsi.fi/uploads/media/vs10XXan.pdf



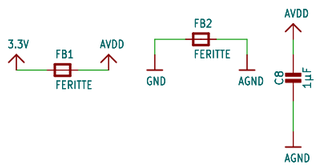
C4,C5 = 2*CL – 2*Cstray
C4,C5 = 2*10pF – 2*5pF
C4,C5 = 10pF

VS1053 oscillator requires 1M resistor for DC bias

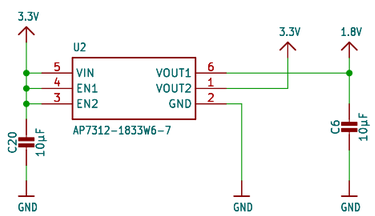
Disable SPI boot with GPIO0 low

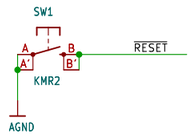


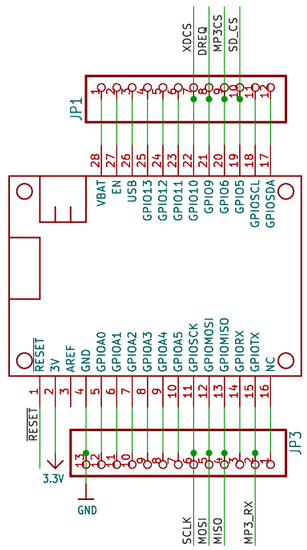
Analog Supply Filtering

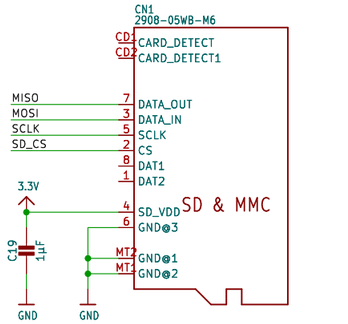


Power Supply/Decoupling









adafruit

${PROJECTNAME}

${CURRENT_DATE} Sheet: ${#}/${##}

Adafruit Industries

Sheet: /
File: working.kicad_sch

Title:

Size: User Date: Rev:

KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1 Id: 1/1